

QUANTITATIVE & QUALITATIVE RESEARCH

Psychologists and other scientists use the scientific approach to study human behavior. Most of their knowledge about the factors that influence human behavior is based on scientific research. In the study of human behavior, both quantitative and qualitative approaches to research are applied, often as complementary approaches to the same topic. This chapter describes the quantitative and qualitative approaches to scientific inquiry, from which much of the knowledge about human behavior is derived.

Quantitative Research

Quantitative research studies human behavior and mental processes by seeking causal relationships among variables, by gathering quantifiable data, and by performing statistical analyses. In quantitative research, the research problem or the hypothesis guides the research process to ensure that the information will be accurate. The research process usually starts with an observation about a particular behavior or event. The researcher then searches for scientific explanations to account for the observation. Through this process, the researcher formulates hypotheses which usually involve a theory. The researcher examines the propositions of the theory through appropriate research designs, samples and sampling technique, and techniques to analyze the data.

Data Selection in Quantitative Research

In quantitative research, the method of selecting the sample is dependent on the purpose and nature of the study. The sample determines the generalizability of the research findings. However, researchers must be careful not to assume that the research findings based on one sample group are generalizable for the larger population. Although it is nice to have a large sample for a research study, a large sample size is not necessary if the participants can be randomly selected and all possible extraneous variables are controlled (Aamodt, 2013).

There are approaches to sampling that are common in quantitative research. In **stratified sampling**, participants are deliberately selected from a variety of levels (strata) or subgroups within the population. **Random sampling** is a sampling technique in which every member of the population has an equal chance of being chosen to participate in the study. The table of random numbers, lottery, and fish bowl are techniques for random sampling. Stratified and random sampling are most likely the methods to ensure that a sample is representative of the larger population. However, there will be times obtaining a stratified and random sample is difficult and thus, convenience or purposive sampling can be used. **Convenience sampling** is a non-probability sampling technique where participants are selected based on the convenience of the researcher. In **purposive sampling**, participants are purposely selected based on particular characteristics. Convenience and purposive sampling are often used for ease of access and availability of samples.

Statistical Analysis

After all the data have been gathered, the results are analyzed statistically. There are two main types of statistical analysis: descriptive and inferential statistics. Descriptive statistics are used to describe the basic features of the data in a study. **Descriptive statistics** such as frequencies, percentages, means, and standard deviations are used to provide basic summaries about the sample and the variables.

Inferential statistics are used to draw inferences or conclusions about associations between variables. There are several kinds of inferential statistics. **Correlation** is used to measure the relationship between two variables. It reveals how closely two variables vary together and so how well one predicts, the other. The **Pearson Product Moment Correlation** is a statistical technique that is so widely used in quantitative research. The result of correlational analysis is a number called a correlation coefficient. The values of a coefficient range from -1 to +1. A positive (+) correlation suggests that as the values

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of one variable increase, so do the values of a second variable increase. A negative (-) correlation suggests that as the values of one variable increase, the values of the second variable decrease.

The **t-test** and **analysis of variance (ANOVA)** are statistical tests that can be used to analyze the differences among means. The t-test is used to compare the means between two groups, whereas the ANOVA is used to compare the means among three or more groups. The one-way analysis of ANOVA determines the statistical differences between the means of two or more independent (unrelated) groups. A two-way ANOVA tests the effect of two independent variables on a dependent variable.

In social sciences, the computation of correlation is usually based on the significant level of 0.05. The software package such as the Statistical Package for the Social Sciences (SPSS) can be used to perform the statistical analysis.

When it comes to data analysis in quantitative studies, reliability and validity are determined for trustworthiness of the overall results. **Reliability** is the extent to which the research study yields consistent results, such as by retesting (replication). Usually, retesting is conducted with different subjects in different situations, to see whether the findings generalize to other subjects and circumstances. **Validity** is used to measure a research method or design. There are two categories of validity: internal and external. **Internal validity** is a measure of how well the study is conducted. It refers to the meaningfulness and confidence in the research results. **External validity** is the degree to which the research results are applicable or can be generalized to other situations or people. Validity is an important part of scientific inquiry. Without a valid research method or design, scientific conclusions cannot be made. In doing so, validity should be based on different types of evidences.

Qualitative Research

Qualitative research studies human behavior by trying to understand the meanings, motives, and beliefs that underlie a person's experiences. 'This approach emphasizes the individual's point of view and the subjective understandings that help account for a person's actions. A major emphasis is on interpretation, made possible by the close acquaintance of the researcher with the details and voices of the participants (Newman & Newman, 2009). Essentially, the most important task of the qualitative researcher is to understand the "insiders' views." The researcher also takes the role of "objective outsider" and relates the interpretive-subjective data to the research purpose and research questions. In qualitative research, the research questions are allowed to evolve, or possibly change, during the study because it is usually focused on exploring phenomena (Christensen, Johnson, & Turner, 2015).

In an attempt to define and describe qualitative research, Yin (2011) proposed five features of qualitative research:

1. Studying the meaning of people's lives, under real-world conditions;
2. Representing the views and perspectives of the participants in a study;
3. Covering the contextual conditions within which people live;
4. Contributing insights into existing or emerging concepts that may help to explain human social behavior; and
5. Striving to use multiple sources of evidence rather than relying on a single source alone.

Qualitative studies pose a special challenge in presenting their data because the data usually include narratives from participants. It is dependent on words rather than numbers for the collected data. The data are likely to be represented in narratives or in data arrays, such as the word tables, hierarchies, matrices, and other types of diagrams (Yin, 2011). It

focuses on self-reports, personal narratives, and expression of ideas, memories, feelings, and thoughts (Myers & Hansen, 2012).

Data Selection in Qualitative Research

In qualitative research, the samples are likely to be chosen in a deliberate manner known as purposive sampling. 'The goal or purpose for selecting the specific study units is to have those that will yield the most relevant and plentiful data (Yin, 2011). The participants should have the knowledge and experience that the researcher requires, be able to reflect and verbalize about the experiences, and be willing to participate in the study. The number of participants or settings in which research occurs is not decided in advance. Typically, the greater the depth and detail in each case, the fewer cases will be included. Additional cases are added as needed until the researcher believes that the variety of perspectives within a setting has been captured and the theoretical insights are well confirmed (Newman & Newman, 2009).

Qualitative studies pose a special challenge in presenting their data because the data usually include narratives from small sample sizes. It relies on words rather than numbers for the data being collected. This approach emphasizes the individual's point of view of the participants and the subjective understandings of the researcher. A major emphasis is on interpretation, made possible by the researcher. Due to small sample sizes and lack of statistical treatments, however, the results of qualitative data cannot be generalized to all populations. Thus, the process of trustworthiness in the research design and method should be properly done. The basis of good qualitative research is the trustworthiness of the data.

There are various approaches to ensure the trustworthiness of the data, but one of the most common approaches is using the criteria of Lincoln and Guba.

Lincoln and Guba (Stahl & King, 2020) suggest that the criteria for trustworthiness in qualitative research are credibility, transferability, dependability, and confirmability. **Credibility** is determining whether the research findings are correct and accurate. Credibility in qualitative research is synonymous with internal validity. The various processes of triangulation is a method in promoting credibility. Triangulating is the use of different sources of information or procedure in establishing patterns. Another method is member checking. This is using informants in validating the analysis of a researcher. It is believed that the use of several methods provide a better understanding of the phenomenon being investigated (Christensen, Johnson, & Turner, 2015). The second criteria for trustworthiness is **transferability**. The transferability of a research finding is the extent to which it can be applied in other contexts and studies. Transferability in qualitative research is synonymous with generalizability or external validity. This is only possible when provided with a substantial interpretation of the data.

The third criteria in trustworthiness is **dependability**. To achieve this, there must be at least two persons to review the contents of the data. It is also necessary for researchers to monitor the influence of their values and beliefs in the interpretation of the data called bracketing. This requires putting aside one's own belief about the phenomenon under investigation or what one already knows about the subject before and throughout the investigation. The last criteria in trustworthiness is **confirmability**. This happens when the qualitative research is confirmed by other researchers.

The primary objective in qualitative research is to learn as much as possible from each participant. In qualitative research, researchers often invite participants to participate in interviews, either interviewed separately or in focus group. Focus group sessions generally last between one to three hours discussing an issue or research topic among six to 12 people with a facilitator. The emphasis is on small-group and in-depth interaction (Christensen, Johnson, & Turner, 2015). Focus group conversations are usually tape-recorded so that researchers can study them in depth (Myers & Hansen, 2012). In a focus group, the facilitator can ask the participants to individually write their responses on a piece of paper to avoid the influence of other participants on their personal or subjective experiences about specific issues. These can then be presented and discussed in the group. Before the focus group discussions, it is important to ask permission from the participants to record the conversation. It is also important to establish rapport with the participants. The interview

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process should be flexible and sensitive to the participant's feelings. Debriefing or emotional processing should be conducted after the discussion to diffuse negative emotions encountered during the process.

Data Analysis in Qualitative Research

Thematic analysis is widely used to analyze data in qualitative research. Thematic analysis is a method that involves reading through a set of data to identify patterns or themes that come up repeatedly such as in transcripts from focus group interviews. There are various approaches to conducting thematic analysis, but one of the most common approaches is the six-step process suggested by Virginia Braun and Victoria Clarke (2006):

Step 1 - Become familiar with the data

After obtaining the data, it is necessary to get a thorough overview of all the data by reading and re-reading through the transcripts and taking initial notes.

Step 2-Generate initial codes

At this stage, individual responses will be analyzed, illustrative quotes will be highlighted, and coded according to similar or common ideas.

Step 3-Search for themes

This stage involves identifying and grouping common ideas into themes. After a process of categorizing and making connections between themes, a higher order of themes will be constructed.

Step 4-Review themes

The emerging higher-order themes will be analyzed, verified, and reviewed to ensure that each theme is clearly represented.

Step 5-Define and name themes

At this stage, each identified higher-order theme will be named. Names need to be concise, punchy, and immediately give the reader a sense of what the theme is about. To visually present the established higher order themes, a table of master themes, sub-themes, and examples of illustrative narratives should be formed. This will encapsulate and summarize the personal or subjective experiences of the research participant.

Step 6-Write-up

Finally, it is important to create a write-up or present the empirical findings of the study.

Narrative analysis is a data analysis technique in qualitative research. It is typically used in studies which have employed narrative inquiry as a qualitative research method. Narrative knowledge is created and constructed through the stories of lived experience and sense-making of the participants. Narrative analysis uses the "story" as the unit of analysis, in contrast to thematic analysis (Lumsden, 2021).

It is important to note, however, research studies do not end with the sheer analysis of their data or the literal presentation of their empirical findings. Good studies must go two further steps-trying to interpret the findings and then drawing some overall conclusion(s) from the study in its entirety (Yin, 2011).

Research Designs, Approaches, and Objectives

Positivism is an approach to quantitative research that depends on empirical scientific evidence, such as experiments. It is an approach that seeks causal relationships among factors, with the objective of trying to predict outcomes. Positivism is based on the premise that knowledge should be gained through observable and measurable evidences. An approach to qualitative research that tries to understand particular situations or phenomena is constructivism. **Constructivism** is an approach that suggests people construct their own understanding of the world through direct or real-life experiences and reflect upon those experiences.

Regardless of whether the quantitative or qualitative approach is utilized, each study needs to be designed. Research designs vary according to the characteristics of the participants, periods of time (time dimension) which data are collected, and the conditions under which the data are collected. The four major research designs are: retrospective cohort study, prospective cohort study, cross-sectional study, and longitudinal study.

In retrospective cohort study, the researcher asks the participants to report on experiences from an earlier time in their lives. The study usually involves taking a cohort of subjects and ask them to recall important periods and events from their past. In a prospective cohort study, the cohort of subjects are identified based on exposure prior to the development of a condition (e.g., disorder, disease) and the outcome is observed over time. Studies that compare people of different ages are called cross-sectional studies. A research study in which the same individuals are observed or studied repeatedly at different points in their lives is called a longitudinal study.

Key Points

- Quantitative research studies human behavior by seeking causal relationships among variables, by gathering quantifiable data, and by performing statistical analyses.
- After all the data have been gathered in quantitative research, the results are statistically analyzed.
- In quantitative research, the two main types of statistical analysis are descriptive and inferential statistics.
- Qualitative research studies human behavior by trying to understand the meanings, motives, and beliefs that underlie a person's experiences.
- Qualitative research relies on words rather than numbers for the data being collected.
- Thematic analysis is widely used to analyze data in qualitative research.

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